7.1.5 Wrap-Up: Cool Down

1. Write an explanation to a classmate who might have missed today's lesson, in your own words, about the Pythagorean Theorem.

Unit 7, Lesson 2: Defining Sine and Cosine

Benchmarks

MA.912.T.1.1 Define trigonometric ratios for acute angles in right triangles.

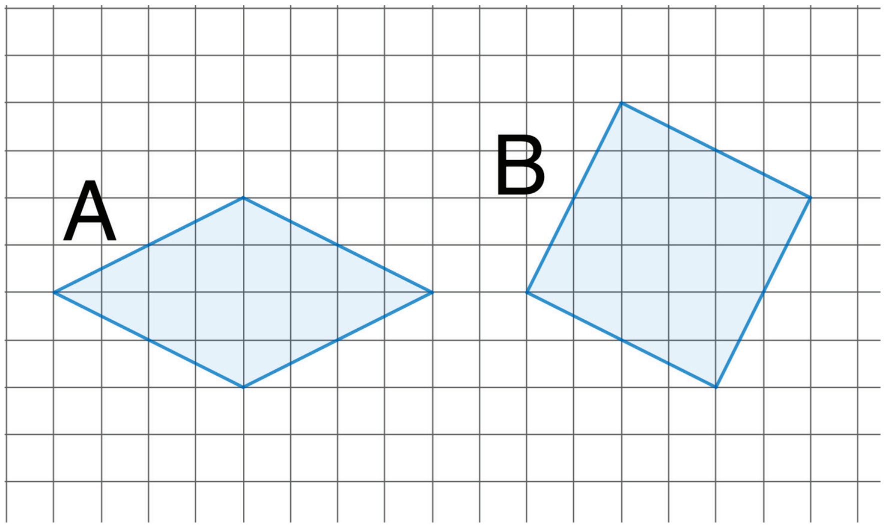
MA.912.T.1.2 Solve mathematical and real-world problems involving right triangles using trigonometric ratios and the Pythagorean Theorem.

Learning Target

☐ I can define sine and cosine using similar triangles.

7.2.1 Warm-Up: Which Region is Larger?

1. Two grids are shown with regions shaded.



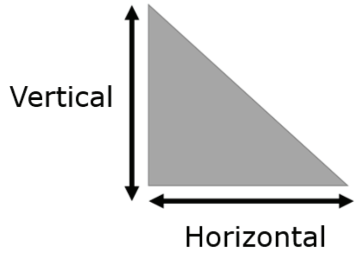
   a. Which shaded region is larger?

   b. Explain your reasoning.

7.2.2 Exploration: Similar Right Triangles

1. Using the space below, create a right triangle that has one angle with a measure of $60°$. Be accurate with measurements.

2. Measure the horizontal and vertical distance of your triangle. Determine the slope of your triangle in centimeters. Round to the nearest tenth.



Vertical	Horizontal	Slope $\left(\frac{\text{vertical}}{\text{horizontal}}\right)$

3. Record the slopes from two of your classmates. *You are the only person who should write in the first two columns of the table.* Have the person initial stating that your slope is correct.

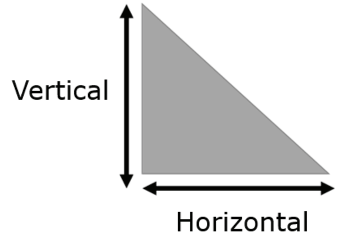
Classmate	Slope	Initials

   a. What do you notice about the slopes?

   b. What do you wonder?

4. Draw a right triangle that has an angle of $45°$. Be accurate with measurements.

5. Measure the horizontal and vertical distance of your triangle. Determine the slope of your triangle in centimeters. Round to the nearest tenth.



Vertical	Horizontal	Slope $\left(\frac{\text{vertical}}{\text{horizontal}}\right)$

6. Record the slopes from two of your classmates. *You are the only person who should write in the second column of the table.* Have the person initial stating that your slope is correct.

Classmate	Slope	Initials

   a. What do you notice about the slopes?

   b. What do you wonder?

7. Ask two classmates for the "Notice and Wonder" what they wrote for both triangles. Record their responses below. *You are the only person who should write in the first three columns of the table.* Have the person initial that it is written correctly.

Classmate	Notice	Wonder	Initials

8. Write a conjecture for each of the right triangles that were drawn.

Angle Measures	Conjecture
$30° - 60° - 90°$	
$45° - 45° - 90°$	

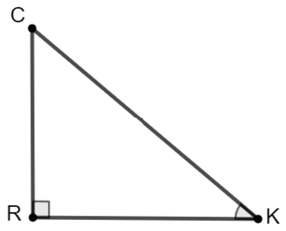
9. The triangles drawn by you and your classmates during the Exploration vary. Using either the triangles explored in Questions 1-3 or Questions 4-6, explain why these triangles are similar.

Right triangles with one other congruent angle are similar because of AA similarity.

7.2.3 Collaboration: Introduction to Trigonometric Ratios

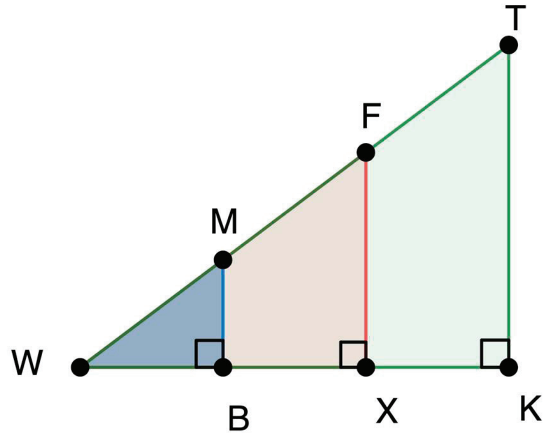
Work with your partner to complete the following.

1. ΔCRK is shown, where $m\angle CRK = 90°$ and $\angle CKR$ is marked as a reference angle.



   a. Draw an arrow from $\angle CKR$ to side $\overline{CR}$. Label that side with an o for opposite.
   b. Label side $\overline{RK}$ with an a for adjacent.
   c. Label the hypotenuse, $\overline{CK}$, with an h for hypotenuse.

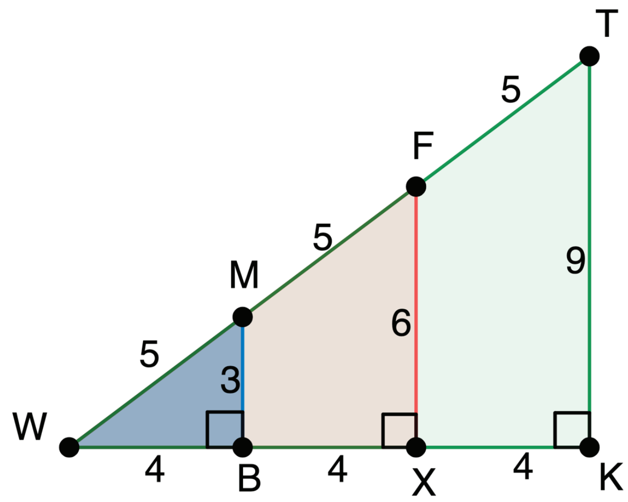
2. The diagram shows three right triangles that share the reference $\angle W$. Explain why ΔWMB, ΔWFX, and ΔWTK are similar.



3. Complete the table.

Triangle	Leg Opposite to $\angle W$	Leg Adjacent to $\angle W$	Hypotenuse
ΔWMB			
ΔWFX			
ΔWTK			

4. The diagram shows measurements for the three triangles.



Complete the table using $\angle W$ as the reference angle.

Triangle	$\frac{\text{opposite}}{\text{hypotenuse}}$	$\frac{\text{adjacent}}{\text{hypotenuse}}$	$\frac{\text{opposite}}{\text{adjacent}}$
ΔWMB			
ΔWFX			
ΔWTK			

7.2.4 Bring It Together: Trigonometric Ratios

Since ΔWMB, ΔWFX, and ΔWTK are similar, the $\frac{\text{opposite}}{\text{hypotenuse}}$ ratios are equivalent to each other. This is also true for the ratios $\frac{\text{adjacent}}{\text{hypotenuse}}$ and $\frac{\text{opposite}}{\text{adjacent}}$. For this reason, it is possible to define these ratios in any right triangle.

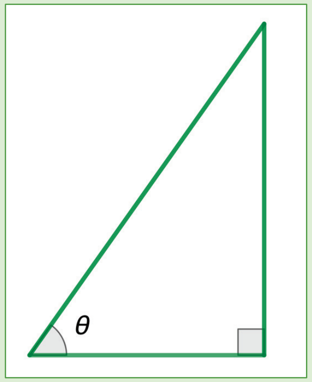
The ratios use the reference angle theta (θ).

Trigonometric Ratios

$$sin(\theta) = \frac{opposite}{hypotenuse}$$

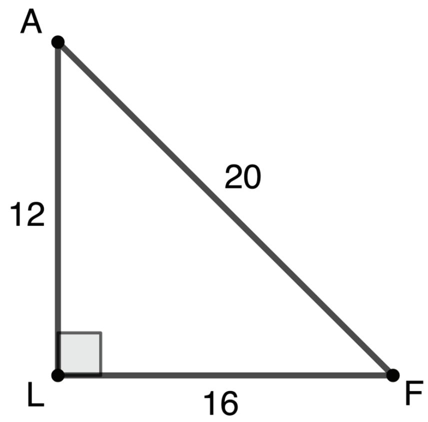
$$cos(\theta) = \frac{adjacent}{hypotenuse}$$

$$tan(\theta) = \frac{opposite}{adjacent}$$



7.2.5 Guided Practice: Trigonometric Ratios

1. ΔFLA is shown. $m\angle ALF = 90°, AL = 12, LF = 16,$ and $AF = 20.$

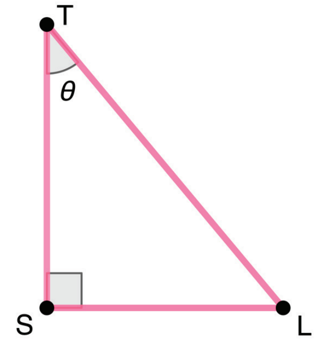


Complete the table using each angle as the reference angle.

Angle	$\frac{\text{opposite}}{\text{hypotenuse}}$	$\frac{\text{adjacent}}{\text{hypotenuse}}$	$\frac{\text{opposite}}{\text{adjacent}}$
$\angle F$			
$\angle A$			

2. ΔTLS is shown where $LS = \sqrt{35}$ and $TS = \sqrt{42}$.

a. Determine the length of $\overline{TL}$.



b. Complete the table for the reference angle θ.

Angle	$\frac{\text{opposite}}{\text{hypotenuse}}$	$\frac{\text{adjacent}}{\text{hypotenuse}}$	$\frac{\text{opposite}}{\text{adjacent}}$
θ			

7.2.6 Wrap-Up: Cool Down

1. Your friend is absent today and will need the notes from class. Write a quick note to explain what was missed in today's lesson.

Unit 7, Lesson 3: Defining Tangent

Benchmarks

MA.912.T.1.1 Define trigonometric ratios for acute angles in right triangles.

MA.912.T.1.2 Solve mathematical and real-world problems involving right triangles using trigonometric ratios and the Pythagorean Theorem.

Learning Target

- ☐ I can use the ratios of sine and cosine to define the tangent ratio.